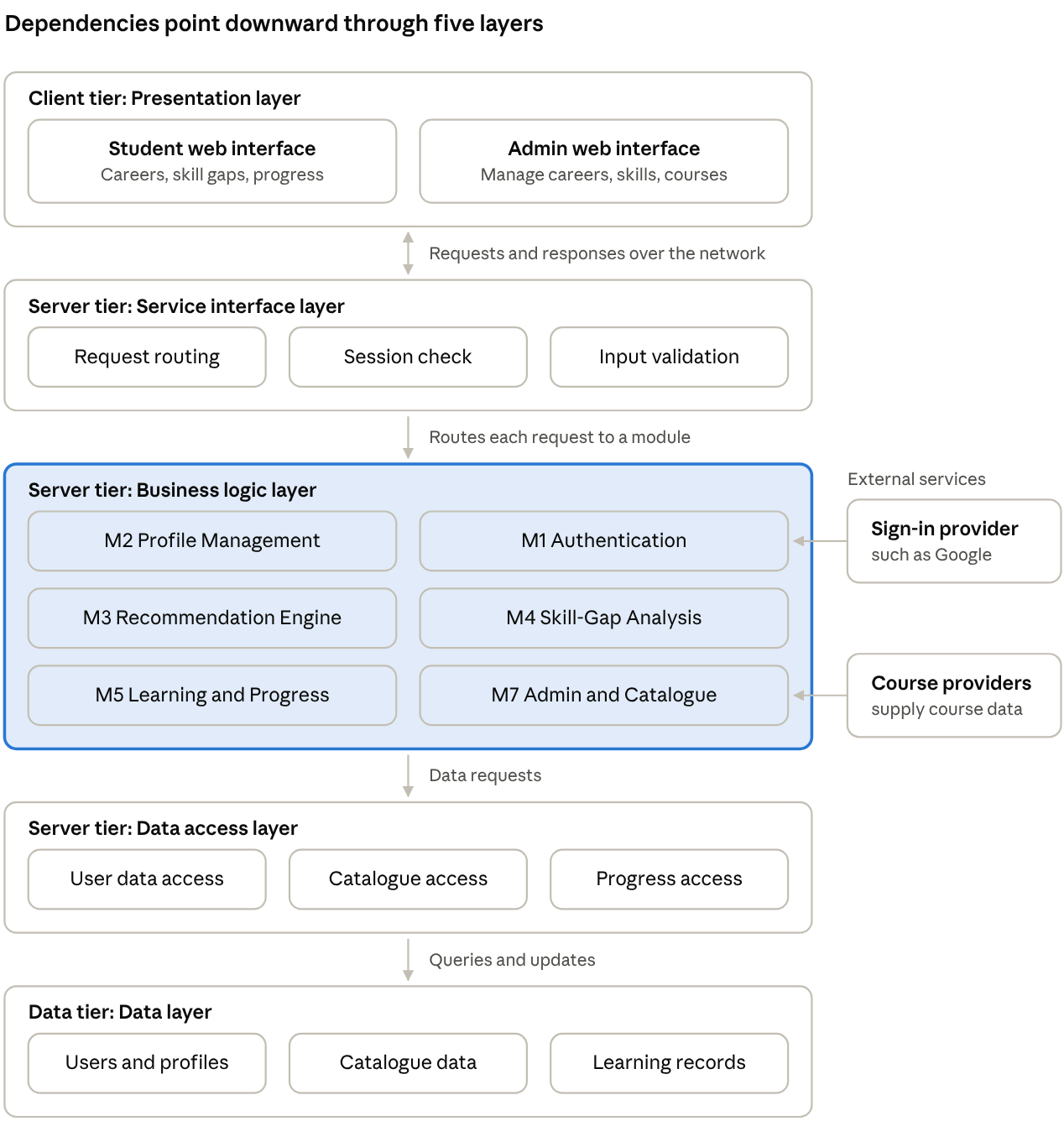
System Architecture Diagram: CareerAI

Session 9 deliverable: a proposed layered, client-server architecture for CareerAI that places the seven modules from the design report into five layers. The technology stack is not specified, so no products or frameworks are named.

Architecture diagram

CareerAI is arranged in five layers across three tiers, and each layer depends only on the layer directly below it. The business logic layer, which holds the modules, is highlighted.



layered client-server architecture · 5 layers, 3 tiers

Layers and responsibilities

Each layer has one job and talks only to the layer below it. All placements are proposed design decisions.

Layer	Tier	Responsibility	Modules and components
Presentation	Client	Responsive dashboard, career search and filtering, feedback, notifications for students and administrators.	M6 User Interface and Feedback
Service interface	Server	Receives requests, checks the session and role, validates input and routes each request to the right module.	Request routing, session check, input validation
Business logic	Server	Authentication, profile handling, career matching with the four recommendation rules, skill-gap analysis, learning recommendations and progress tracking, catalogue administration.	M1, M2, M3, M4, M5, M7
Data access	Server	Gives modules one controlled way to read and write stored data, so modules never touch storage directly.	User data, catalogue and progress access components
Data	Data	Stores user accounts and profiles, careers, skills and courses, and learning records.	Users and profiles, catalogue data, learning records

External services sit outside the layers. A third-party sign-in provider, such as Google, supports M1, and course providers supply the course and certification information that administrators maintain through M7.

Architecture notes

Client-server. The browser-based client handles presentation only, and the server tier holds all business rules and data access. This keeps one copy of the matching and gap logic for every device size, and lets the server meet the five-second recommendation target and protect user data in one place.

Layering. Each layer depends only on the one below it, which gives low coupling between layers and lets a layer change without rewriting the others. For example, the catalogue storage can change without touching M3.

Microservices (conceptual). The modules already have separate responsibilities and narrow interfaces, so M3, M4 and M5 could later be split into independent services if load or team size requires it. This is a future option, not part of the current design, which stays a single layered application.

Scope. The architecture covers the responsive web platform only. Native mobile apps, applicant tracking system integration, video counselling and payment processing are outside the current phase.